```python
import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from sklearn.linear_model import LogisticRegression
from sklearn.metrics import (
    classification_report, confusion_matrix,
    roc_auc_score, roc_curve,
    precision_recall_curve, f1_score
)

%matplotlib inline
```

```python
train = pd.read_csv('../data/processed/training_lr_mlp.csv')
test  = pd.read_csv('../data/processed/test_lr_mlp.csv')

X_train = train.drop(columns=['label'])
y_train = train['label']
X_test  = test.drop(columns=['label'])
y_test  = test['label']

print(f'Train: {X_train.shape}, Test: {X_test.shape}')
print(f'Class balance (train) — 0: {(y_train==0).sum()}, 1:
{(y_train==1).sum()}')
```

```
Train: (107740, 58), Test: (55945, 58)
Class balance (train) — 0: 51890, 1: 55850
```

```python
base_model = LogisticRegression(
    solver='saga',
    penalty='l2',
    C=1.0,
    max_iter=2000,
    random_state=42
)
```

Hyperparameter Tuning

```python
from sklearn.model_selection import RandomizedSearchCV
from scipy.stats import loguniform, uniform

param_dist = [
    {
        'C': loguniform(1e-3, 1e3),
        'penalty': ['l1', 'l2'],
        'solver': ['saga'],
        'class_weight': [None, 'balanced'],
        'max_iter': [2000],
    },
    {
```

```
        'C': loguniform(1e-3, 1e3),
        'penalty': ['elasticnet'],
        'solver': ['saga'],
        'l1_ratio': uniform(0.1, 0.8),
        'class_weight': [None, 'balanced'],
        'max_iter': [2000],
    },
]

search = RandomizedSearchCV(
    base_model,
    param_distributions=param_dist,
    n_iter=30,
    scoring='roc_auc',
    cv=3,
    n_jobs=-1,
    random_state=42,
    verbose=1
)

search.fit(X_train, y_train)

print(f'Best ROC-AUC (CV): {search.best_score_:.4f}')
print(f'Best params:       {search.best_params_}')
```

```
Fitting 3 folds for each of 30 candidates, totalling 90 fits

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Best ROC-AUC (CV): 0.9435
Best params:       {'C': np.float64(897.9855655182433),
'class_weight': None, 'l1_ratio': np.float64(0.3433937943676302),
'max_iter': 2000, 'penalty': 'elasticnet', 'solver': 'saga'}

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```

```
# hard-coding the optimal parameters from previous hyperparameter run
to prevent running it again.
best_params = {'C': np.float64(897.9855655182433), 'class_weight':
None, 'l1_ratio': np.float64(0.3433937943676302), 'max_iter': 2000,
'penalty': 'elasticnet', 'solver': 'saga'}

#model = search.best_estimator_
model = LogisticRegression(**best_params, random_state=42)

model.fit(X_train, y_train)
```

```
/Users/firefly/Desktop/_c11/final_project/ml-network-intrusion/.venv/
lib/python3.12/site-packages/sklearn/linear_model/_logistic.py:1135:
FutureWarning: 'penalty' was deprecated in version 1.8 and will be
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```
LogisticRegression(C=np.float64(897.9855655182433),
                   l1_ratio=np.float64(0.3433937943676302),
max_iter=2000,
                   penalty='elasticnet', random_state=42,
solver='saga')
```

Base Evaluation, threshold = 0.5

```
y_prob = model.predict_proba(X_test)[:, 1]
y_pred = (y_prob >= 0.5).astype(int)

print(classification_report(y_test, y_pred, target_names=['Normal',
'Attack']))
print(f'ROC-AUC: {roc_auc_score(y_test, y_prob):.4f}')
```

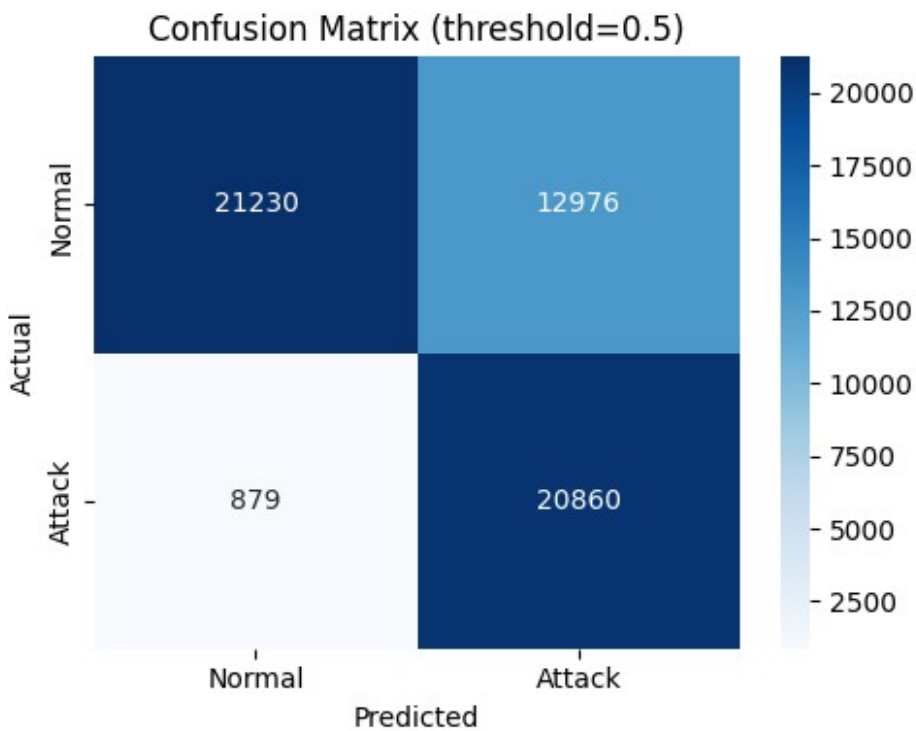
```
              precision    recall  f1-score   support

      Normal       0.96      0.62      0.75     34206
      Attack       0.62      0.96      0.75     21739

    accuracy                           0.75     55945
   macro avg       0.79      0.79      0.75     55945
weighted avg       0.83      0.75      0.75     55945

ROC-AUC: 0.9217
```

```
cm = confusion_matrix(y_test, y_pred)
fig, ax = plt.subplots(figsize=(5, 4))
sns.heatmap(cm, annot=True, fmt='d', cmap='Blues', ax=ax,
            xticklabels=['Normal', 'Attack'], yticklabels=['Normal',
'Attack'])
ax.set_ylabel('Actual')
ax.set_xlabel('Predicted')
ax.set_title('Confusion Matrix (threshold=0.5)')
plt.tight_layout()
plt.show()
```



Threshold Tuning

In cybersecurity, false negatives (missed attacks) are more costly than false positives. We find the threshold that maximises F1, then also examine the recall-precision tradeoff.

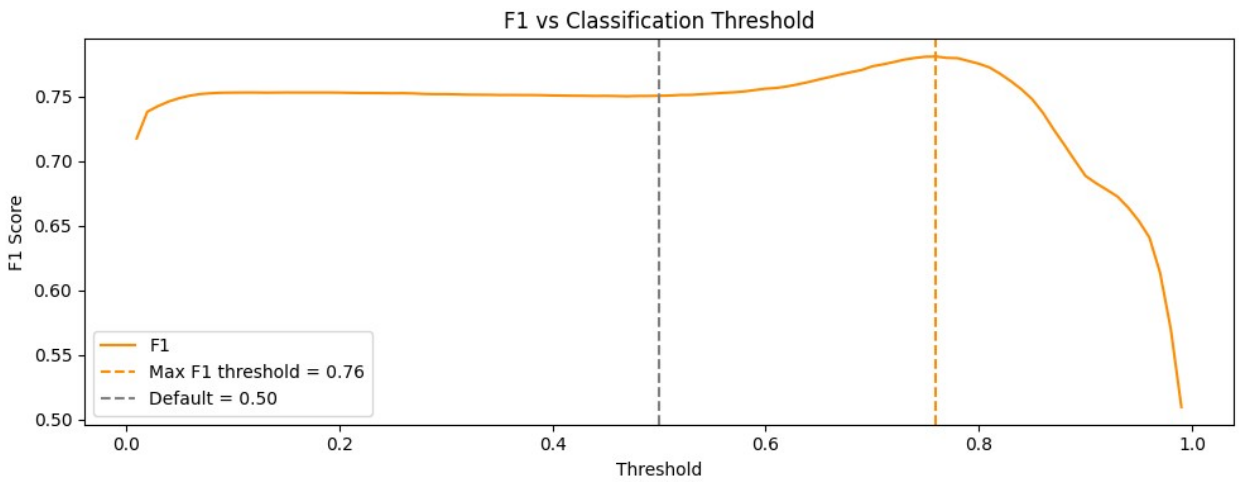
```
thresholds = np.arange(0.01, 1.0, 0.01)
f1_scores  = [f1_score(y_test, (y_prob >= t).astype(int)) for t in
thresholds]

best_f1_threshold = thresholds[np.argmax(f1_scores)]

print(f'Optimal threshold (max F1): {best_f1_threshold:.2f}  — F1:
{max(f1_scores):.4f}')
print(f'Default threshold (0.5):           — F1: {f1_scores[49]:.4f}')
```

```
fig, ax = plt.subplots(figsize=(10, 4))
ax.plot(thresholds, f1_scores, color='darkorange', label='F1')
ax.axvline(best_f1_threshold, color='darkorange', linestyle='--',
label=f'Max F1 threshold = {best_f1_threshold:.2f}')
ax.axvline(0.5, color='gray', linestyle='--', label='Default = 0.50')
ax.set_xlabel('Threshold')
ax.set_ylabel('F1 Score')
ax.set_title('F1 vs Classification Threshold')
ax.legend()
plt.tight_layout()
plt.show()
```

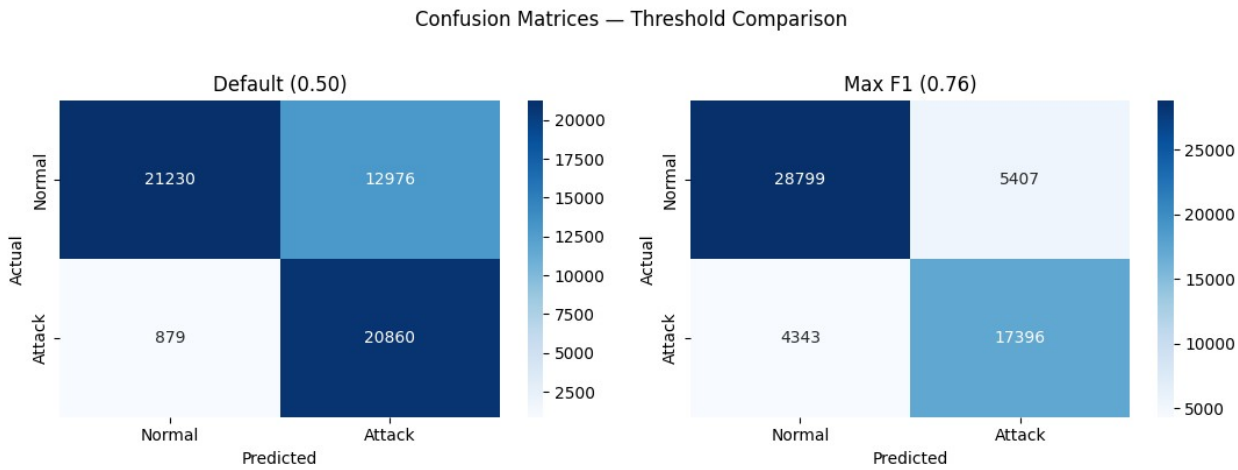
```
Optimal threshold (max F1): 0.76  — F1: 0.7811
Default threshold (0.5):           — F1: 0.7507
```




```
fig, axes = plt.subplots(1, 2, figsize=(11, 4))
for ax, (t, title) in zip(axes, [
    (0.5, 'Default (0.50)'),
    (best_f1_threshold, f'Max F1 ({best_f1_threshold:.2f})'),
]):
    preds = (y_prob >= t).astype(int)
    sns.heatmap(confusion_matrix(y_test, preds), annot=True, fmt='d',
cmap='Blues', ax=ax,
                xticklabels=['Normal', 'Attack'],
yticklabels=['Normal', 'Attack'])
    ax.set_ylabel('Actual')
    ax.set_xlabel('Predicted')
    ax.set_title(title)
plt.suptitle('Confusion Matrices — Threshold Comparison', y=1.02)
plt.tight_layout()
plt.show()

print(f'\nChosen threshold: {best_f1_threshold:.2f} (max F1)')
```

```
print(classification_report(y_test, (y_prob >=
best_f1_threshold).astype(int), target_names=['Normal', 'Attack']))
```




```
Chosen threshold: 0.76 (max F1)
              precision    recall  f1-score   support

      Normal       0.87      0.84      0.86     34206
      Attack       0.76      0.80      0.78     21739

    accuracy                           0.83     55945
   macro avg       0.82      0.82      0.82     55945
weighted avg       0.83      0.83      0.83     55945


y_pred_tuned = (y_prob >= best_f1_threshold).astype(int)
```

Per-Attack-Category Breakdown

Join predictions against the raw test data to evaluate performance per attack type.

```
raw_test = pd.read_csv('../data/raw/UNSW_NB15_testing-set.csv')

raw_test = raw_test.drop_duplicates(subset=[c for c in
raw_test.columns if c != 'id']).reset_index(drop=True)

results = raw_test[['attack_cat']].copy()
results['y_true'] = y_test.values
results['y_pred'] = y_pred_tuned
results['y_prob'] = y_prob

results.head()
```

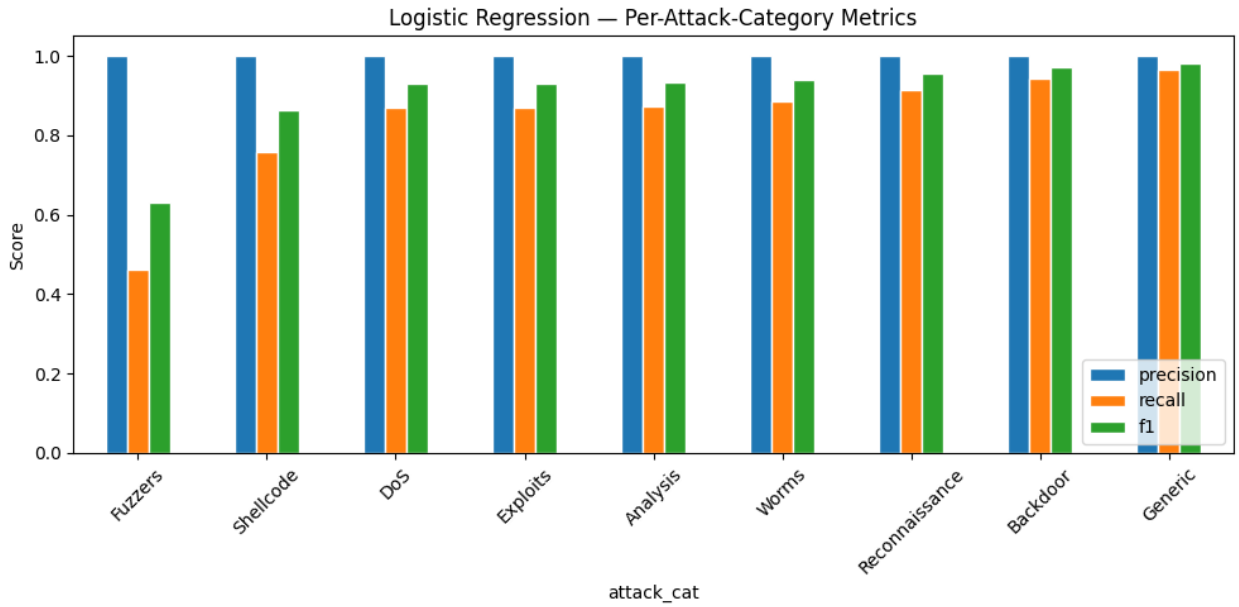
```
  attack_cat  y_true  y_pred    y_prob
0     Normal       0       1  0.880819
1     Normal       0       1  0.829518
2     Normal       0       1  0.843379
3     Normal       0       1  0.876930
4     Normal       0       1  0.814249
```

```
def attack_metrics(group):
    tp = ((group['y_true'] == 1) & (group['y_pred'] == 1)).sum()
    fn = ((group['y_true'] == 1) & (group['y_pred'] == 0)).sum()
    fp = ((group['y_true'] == 0) & (group['y_pred'] == 1)).sum()
    tn = ((group['y_true'] == 0) & (group['y_pred'] == 0)).sum()
    precision = tp / (tp + fp) if (tp + fp) > 0 else 0
    recall = tp / (tp + fn) if (tp + fn) > 0 else 0
    f1 = 2 * precision * recall / (precision + recall) if (precision +
recall) > 0 else 0
    return pd.Series({'count': len(group), 'precision':
round(precision, 4),
                      'recall': round(recall, 4), 'f1': round(f1, 4)})

attack_results = results[results['attack_cat'] != 'Normal']
breakdown = attack_results.groupby('attack_cat').apply(attack_metrics,
include_groups=False).sort_values('recall')
print(breakdown.to_string())
```

```
                  count  precision  recall      f1
attack_cat
Fuzzers          4838.0        1.0  0.4618  0.6318
Shellcode         378.0        1.0  0.7593  0.8632
DoS              1718.0        1.0  0.8696  0.9303
Exploits         7609.0        1.0  0.8705  0.9308
Analysis          446.0        1.0  0.8722  0.9317
Worms              44.0        1.0  0.8864  0.9398
Reconnaissance   2703.0        1.0  0.9145  0.9554
Backdoor          346.0        1.0  0.9422  0.9702
Generic          3657.0        1.0  0.9655  0.9825
```

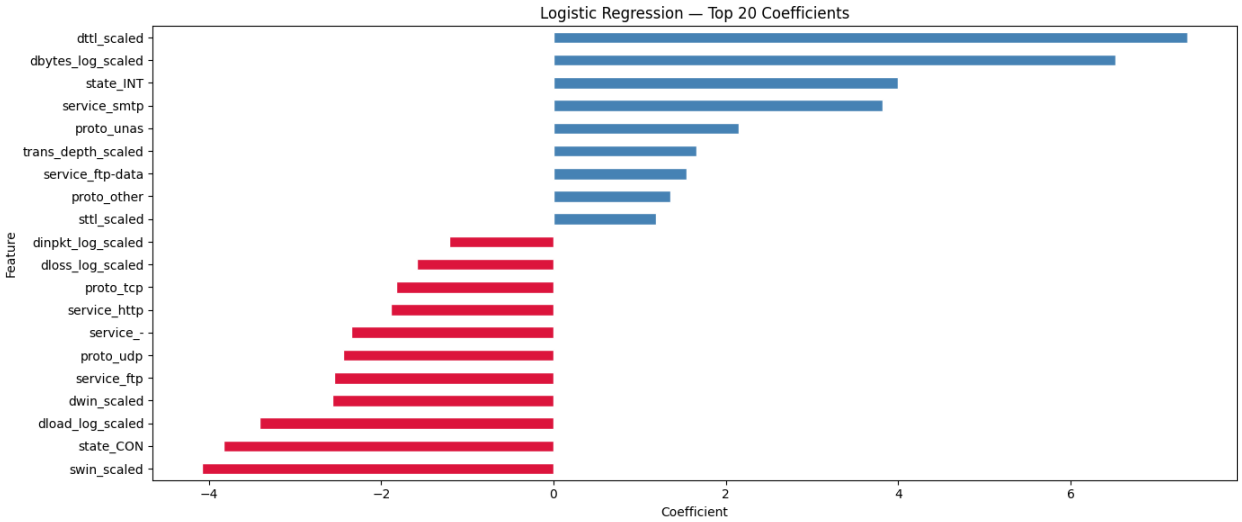
```
fig, ax = plt.subplots(figsize=(10, 5))
breakdown[['precision', 'recall', 'f1']].plot(kind='bar', ax=ax,
edgecolor='white')
ax.set_title('Logistic Regression — Per-Attack-Category Metrics')
ax.set_ylabel('Score')
ax.set_ylim(0, 1.05)
ax.tick_params(axis='x', rotation=45)
ax.legend(loc='lower right')
plt.tight_layout()
plt.show()
```



Feature Importance

```
coefficients = pd.Series(model.coef_[0], index=X_train.columns)
top_features =
coefficients.abs().sort_values(ascending=False).head(20).index
top_coefficients = coefficients.loc[top_features].sort_values()
colors = ['crimson' if value < 0 else 'steelblue' for value in
top_coefficients]

fig, ax = plt.subplots(figsize=(14, 6))
top_coefficients.plot(kind='barh', ax=ax, color=colors,
edgecolor='white')
ax.set_title('Logistic Regression — Top 20 Coefficients')
ax.set_xlabel('Coefficient')
ax.set_ylabel('Feature')
plt.tight_layout()
plt.show()
```


Logistic Regression — Top 20 Coefficients
dttl_scaled
dbytes_log_scaled
state_INT
service_smtp
proto_unas
trans_depth_scaled
service_ftp-data
proto_other
sttl_scaled
dinpkt_log_scaled
dloss_log_scaled
proto_tcp
service_http
service_-
proto_udp
service_ftp
dwin_scaled
dload_log_scaled
state_CON
swin_scaled
Feature
−4
−2
0
2
4
6
Coefficient